RAJALAKSHMI ENGINEERING COLLEGE

PHYSICS-GUIDED MACHINE LEARNING AND FAULT SEVERITY MONITORING FRAMEWORK FOR A PEMFC

Final Project Presentation • Academic Year 2026–2027

PROJECT TEAM

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GUIDE & DEPARTMENT

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OVERVIEW

Presentation Roadmap

From physics understanding → fault representation → classification → severity monitoring

01
Problem & Objective

02
Physics-Guided Workflow

03
Sensitivity & Fault Representation

04
Fault Dataset Generation & Qualification

05
Fault Classification & Feature Architecture

06
Explainability & Physical Consistency

07
Experimental / Blind Assessment

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Conclusions & Limitations

Problem Statement & Research Objective

Physics-guided fault diagnosis for interpretable PEMFC fault type and severity identification

THE PROBLEM

Conventional data-driven fault diagnosis can achieve high
classification accuracy but may lack physical interpretability
and defensible fault-severity representation.

CORE OBJECTIVE

Develop a physics-guided machine-learning framework
for PEMFC fault diagnosis and severity identification using
physically meaningful fault representations and PEMFC-informed features.

STUDENT API OPERATING INPUTS

Current • anode pressure • cathode pressure • anode RH
• cathode RH • sink temperature

PHYSICS-GUIDED DIAGNOSTIC VARIABLES

Cell voltage • stack voltage • temperature
• effective membrane hydration • minimum H_2 •
minimum O_2

RESEARCH EMPHASIS

Fault representation → feature architecture → classification
→ severity interpretation

Reference PEMFC & Physics Boundary

Frozen reference model supplies operating states and physically interpretable variables

INPUT VECTOR

I
p_a, p_c
RH_a, RH_c
T_s

FROZEN PEMFC

25 cm^2 active area
6-cell stack
MATLAB/API execution
Solver convergence logging

OUTPUTS

Vcell
Vstack
T
λeff
$Cmin,O_2$
$Cmin,H_2$

Physics boundary: Frozen 25 cm^2 / 6-cell PEMFC model provides the operating states and physically interpretable variables used for fault representation and diagnosis.

METHOD

End-to-End Methodology

Controlled physics analysis → fault dataset → classification → severity and assessment

1 **Frozen 25 cm² Reference**
Student API + fixed configuration

2 **Physics Sensitivity**
Current / pressure / RH / temperature response

3 **Fault Representation**
F0 Normal + F1 / F2 / F3 fault families

4 **Fault Dataset Generation**
100 qualified cases • 10 current levels

5 **Feature Engineering**
Measured + derived + physics-guided features

6 **ML Classification**
Random Forest / SVM / ANN

7 **Severity Identification**
Fault type + mild / moderate / severe

8 **Explainability & Assessment**
Feature importance + physical consistency + 25 / 100 cm² assessment

S3 Controlled Sensitivity Study

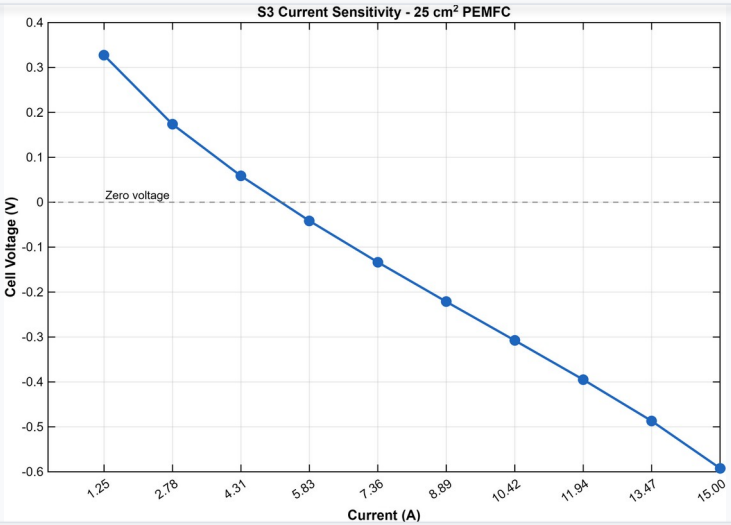
One-factor-at-a-time sensitivity establishes the physical reference trends used to interpret fault signatures.

6	7	2026	25 cm²
factors	points / factor*	DOE seed	reference stack

TESTED ENVELOPE

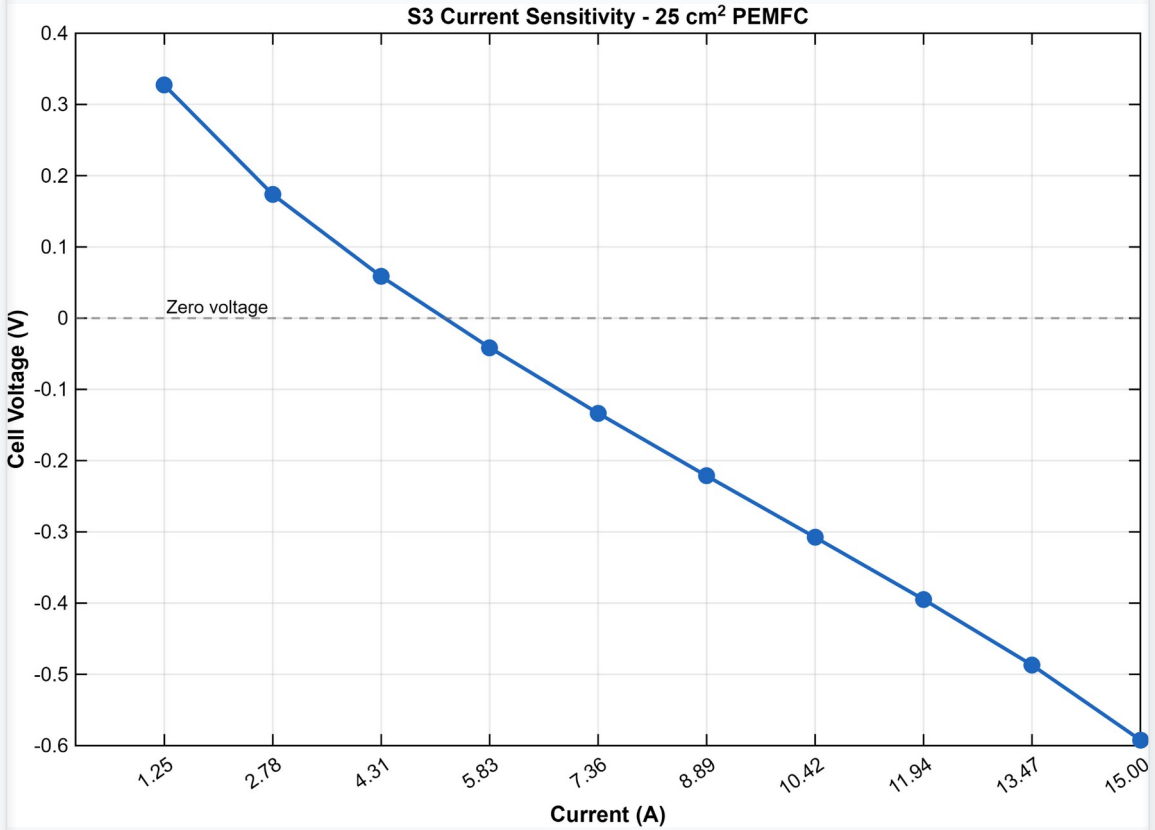
Current: 1.25–15 A
Anode / cathode pressure: 100–220 kPa
Anode / cathode RH: 0.30–1.00
Sink temperature: 333–353 K

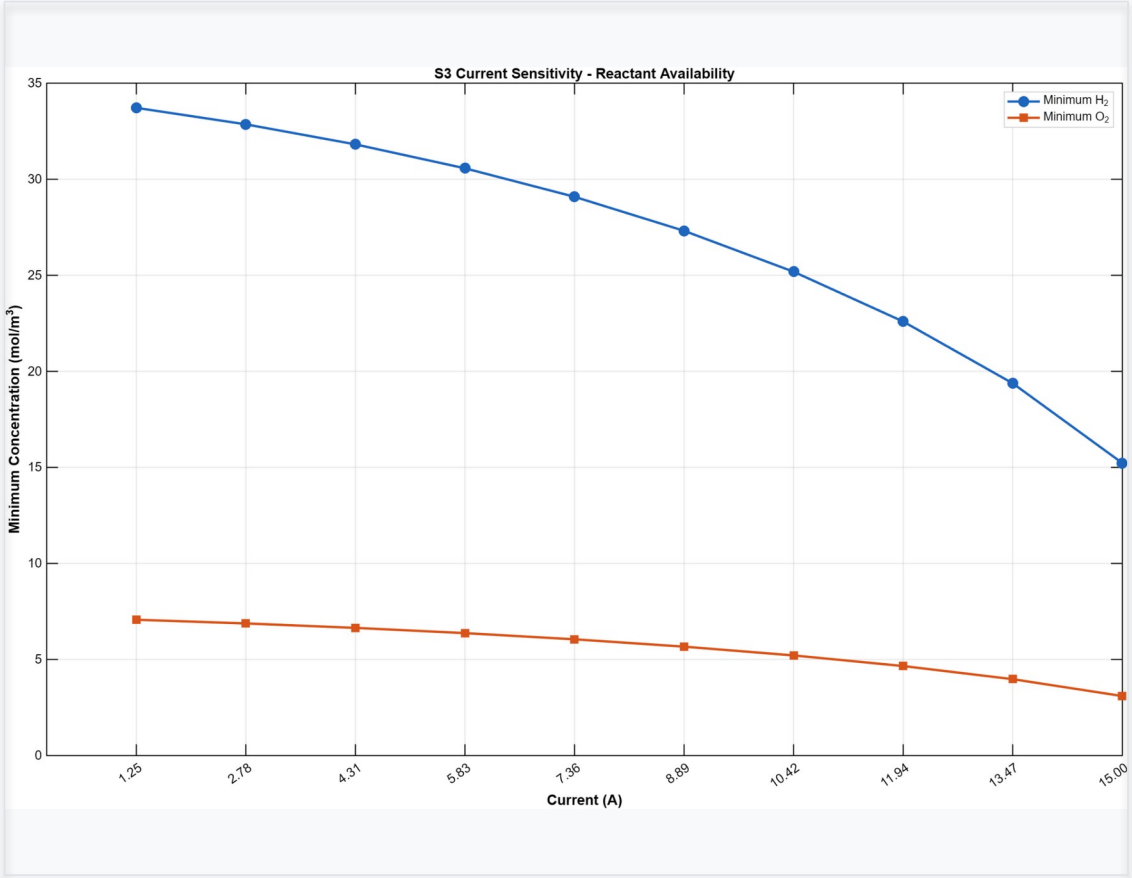
*10 current points; 7 points for the other factors.



S3 Physical Response — Current & Reactants

These trends provide the physical baseline for interpreting F1–F3 fault responses.



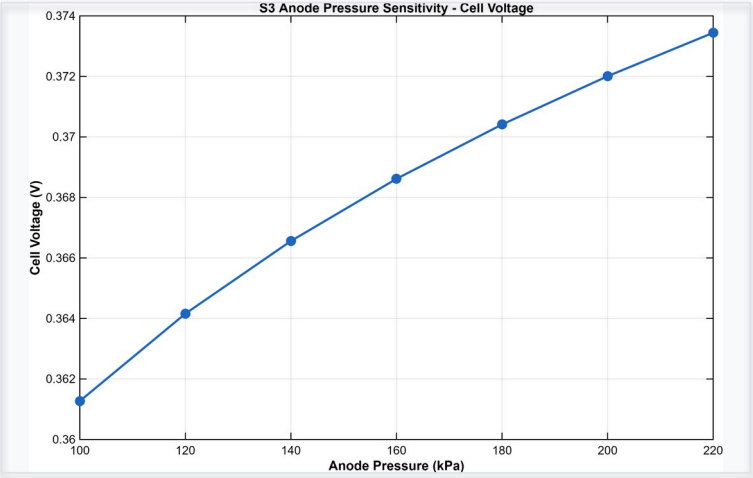


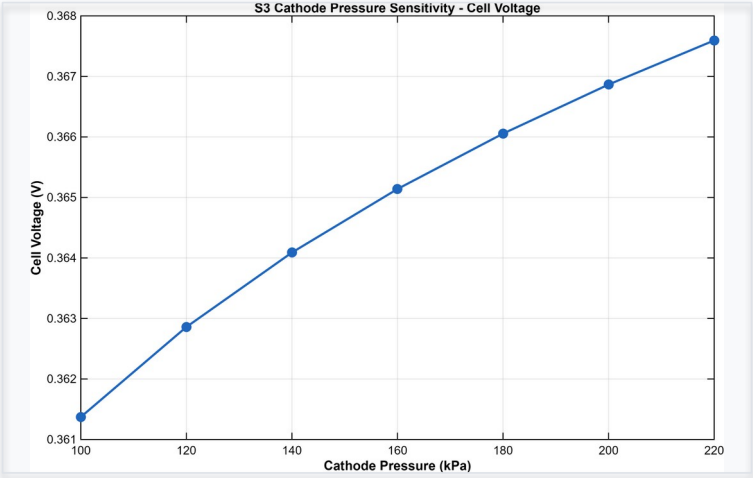
PHYSICAL BASELINE

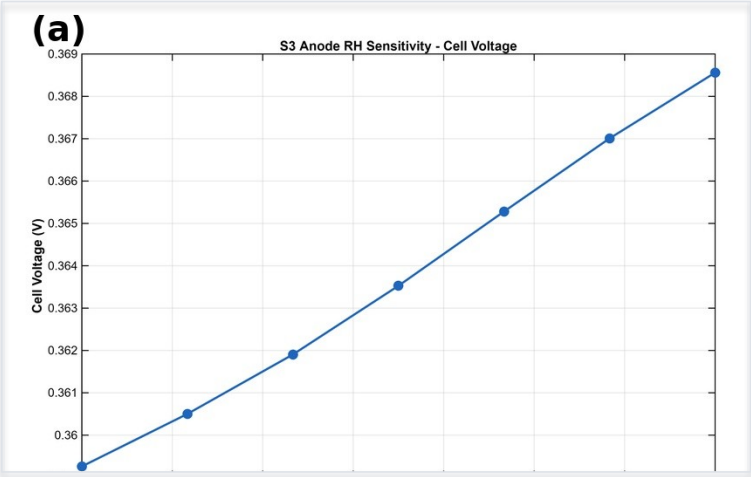
These trends provide the physical baseline for interpreting F1–F3 fault responses. The high-current convergence boundary is retained as a numerical limitation of the frozen model,
not silently removed data.

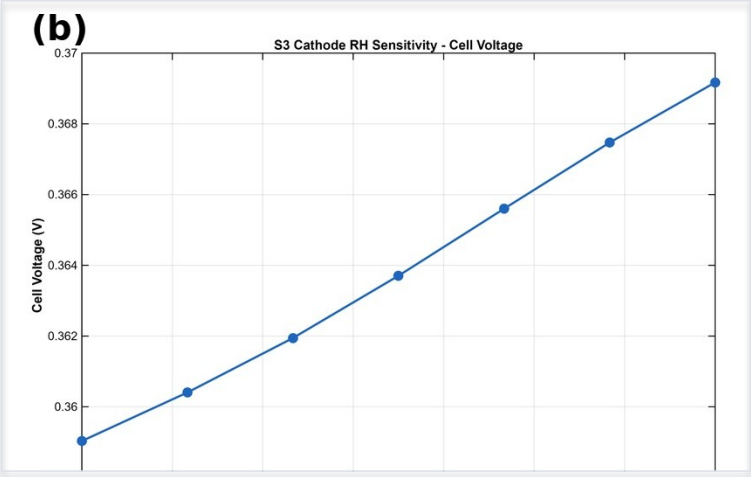
S3 Physical Response — Pressure & Humidity

Pressure and humidity provide clear reference trends for later surrogate checks









PHYSICAL REFERENCE

These physical trends are used as reference behaviour when interpreting the physics-guided features and fault signatures.

S110 Fault Dataset — Generation & Qualification

Controlled fault representation with normal, mild, moderate and severe cases

100	10	10	100 / 100
total fault cases	fault/severity combinations	current operating points	converged & finite cases

DATASET STRUCTURE

F0 Normal — 10 cases
F1 Drying — 30 cases
F2 Oxygen starvation — 30 cases
F3 Over-hydration / transport — 30 cases

SEVERITY DESIGN

Every fault family contains mild / moderate / severe cases.

OPERATING GRID

Current: 0.5–2.0 A • 10 levels
Fault conditions are controlled through RH or cathode pressure.

ML RESULTS

Baseline Fault Classification Models

Controlled internal split using the 24-feature Model C architecture

Model	Test Accuracy	Macro-F1
Random Forest	1.000	1.000
SVM	0.7059	0.5606
ANN	1.000	1.000

70 / 13 / 17%

train / val / test

24

Model-C features

FEATURE SET

Measured + derived + physics-guided variables; frozen Model-C architecture.

KEY TAKEAWAY

RF and ANN achieved perfect internal test
classification, while SVM showed weaker
generalization and class confusion.

Physics-Guided Feature Architecture

Three feature architectures test whether physics-informed variables add diagnostic value

MODEL A — EXPERIMENTAL / MEASURED

14 features
Directly measured operating / response
variables.

MODEL B — MEASURED + DERIVED

21 features
Adds derived diagnostic quantities.

MODEL C — PHYSICS-GUIDED

24 features
Adds mechanism-based PEMFC
variables.

Architecture	RF Test Accuracy	RF Macro-F1
Model A	1.000	1.000
Model B	1.000	1.000
Model C	1.000	1.000

SCIENTIFIC CONCLUSION

Physics-guided features did not increase internal accuracy, but they improved physical interpretability
and provide a mechanism-based representation of the
fault signatures.

Explainability & Physical Consistency

Feature importance connects the classifier to physically meaningful PEMFC fault signatures

Physics-Guided Feature	Importance
LambdaEffective	20.07%
RHCathode	19.00%
RHAnode	13.73%
pCathode	12.88%
H_2/O_2 concentration ratio	10.14%
Minimum O_2	9.71%
Minimum H_2	9.29%

MODEL-SPECIFIC PHYSICS IMPORTANCE

Physics-guided features contributed approximately 20.29% of total
model-specific importance.

FAULT SIGNATURES

F1: hydration ↓ / voltage ↓
F2: minimum O_2 ↓ / voltage ↓
F3: hydration ↑ / voltage ↑

INTERPRETATION BOUNDARY

Feature importance indicates model dependence, not causal proof.

Experimental & Cross-Scale Assessment

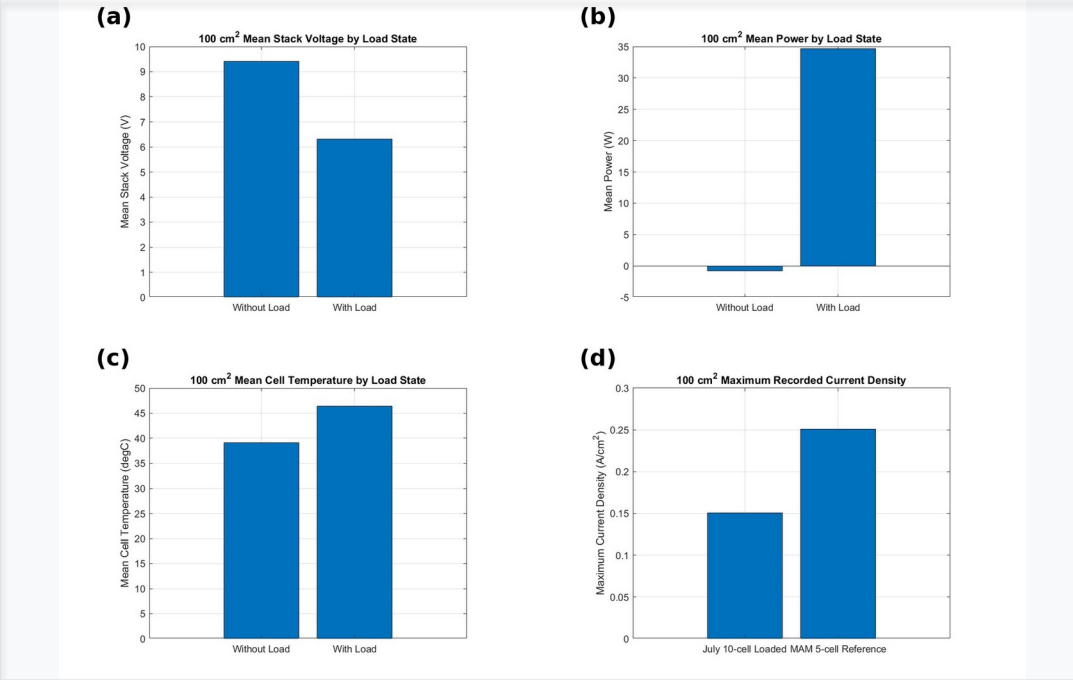
Physical assessment is separated from direct classifier-transfer claims

25 cm² EXPERIMENTAL ASSESSMENT

- 2201 experimental samples
- Current: −0.12 to 30.13 A
- Normal-domain experimental samples: 138
- Normal-domain fraction: 6.27%
- Independent experimental fault labels: 0

CLAIM BOUNDARY

No direct experimental fault-classifier validation is claimed because
independent experimental fault labels are unavailable.



100 cm² EXTERNAL PACKAGE

6801 July 10-cell samples • Loaded mean power
34.664 W
• Maximum power 60.42 W

Fault-Severity Monitoring Framework

Controlled fault families are represented at mild, moderate and severe levels

F1 • MEMBRANE DRYING

Severity controlled through reduced RH:
Mild 0.45 • Moderate 0.30 • Severe 0.10

Expected signature: lower hydration / lower voltage.

F2 • OXYGEN STARVATION

Severity controlled through cathode pressure:
Mild 95 kPa • Moderate 75 kPa • Severe 60 kPa

Tracked using minimum O_2 and cell voltage.

F3 • OVER-HYDRATION / TRANSPORT

Severity controlled through increased RH:
Mild 0.60 • Moderate 0.70 • Severe 0.90

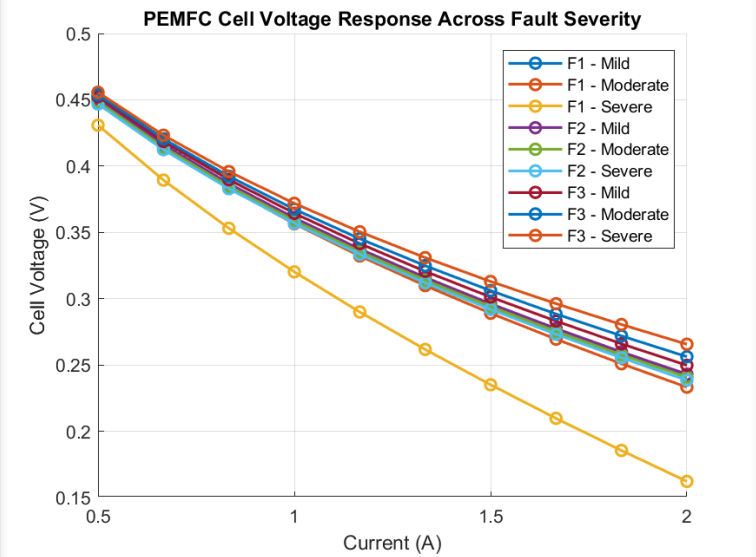
Tracked using voltage and hydration response.

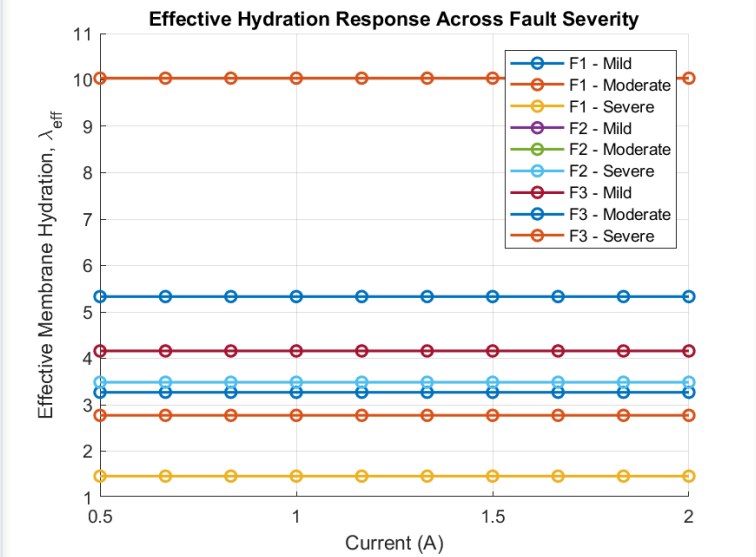
S110 STRUCTURE

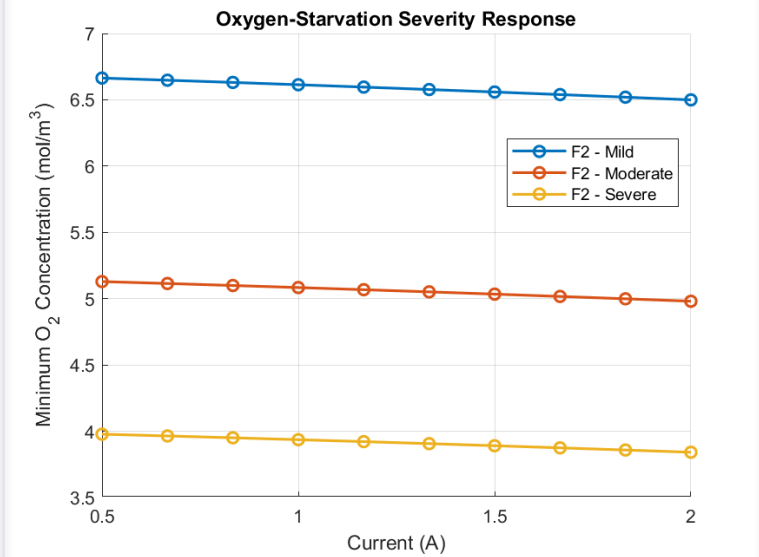
100 cases • F0 normal + F1/F2/F3 faults • 10 current levels from 0.5–2.0 A • fixed anode pressure 101.325 kPa
• fixed sink/initial temperature 328.15 K • convergence and finite status retained.

S15 Fault-Severity Responses

Physical response plots support severity interpretation without claiming classifier accuracy







INTERPRETATION

Cell voltage, effective hydration and minimum O_2 provide physically interpretable severity responses. No classifier accuracy is claimed from these plots.

RESULTS

Key Results at a Glance

Classification performance is strong internally; physics-guided features add interpretability rather than an accuracy gain

1.000 RF test accuracy	**1.000** ANN test accuracy	**0.7059** SVM test accuracy
100 qualified fault cases	**0** adjacent-severity errors	**3** fault families

CENTRAL MESSAGE

The framework combines controlled fault representation, physics-guided features, baseline ML classification, severity identification and physically
interpretable fault responses.

100 cm² Assessment — Scientific Boundary

The frozen classifier is not transferred when required Model-C inputs are unavailable

25 cm² DEVELOPMENT

Frozen reference model
Fault dataset + feature architecture

MODEL FREEZE

No retraining
No feature changes

100 cm² PACKAGE

Physical / operational assessment of the external population

TRANSFER STATUS

Frozen classifier transfer: NOT EXECUTED

Required Model-C variables were unavailable in the 100 cm² package.

INTEGRITY CONTROLS

No retraining
No feature modification
No threshold modification
No hyperparameter modification
No artificial features

Conclusions & Future Work

A traceable path from PEMFC physics to fault diagnosis and severity interpretation

CONCLUSIONS

• A physics-guided PEMFC fault-diagnosis framework was developed around a frozen 25 cm^2 / 6-cell reference model.
• Three fault families were represented: membrane drying, oxygen starvation
and effective over-hydration / transport impairment.
• A qualified 100-case fault dataset covering normal, mild, moderate and
severe conditions was generated.
• RF and ANN achieved 1.000 internal test accuracy and Macro-F1.
• Physics-guided features did not produce an accuracy gain, but provided
physically interpretable diagnostic information.
• Severity identification achieved zero adjacent-severity errors on the evaluated test cases.
• Direct 100 cm^2 classifier transfer was not executable because required
frozen features were unavailable.

FUTURE WORK

• Increase fault-dataset diversity and operating envelope.
• Obtain externally measured fault-labelled datasets.
• Validate the classifier on a complete 100 cm^2 Model-C-compatible
measurement set.
• Investigate robustness under unseen operating conditions.
• Extend the framework toward experimentally validated online fault
diagnosis.

THANK YOU

Questions & Discussion

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